Integer Array Practice Questions

Write a function for each of the following questions. The function takes one or more arrays of integers and **returns** the answer. The function must return the result, not print it.

Quick reminder: `return result;` is correct. `print(result)` inside the function is not.

1

Count how many numbers are odd and how many are even. Return both as [odd, even].

Input to the function: `[3, 8, 5, 12, 7]`
Return from the function: `[3, 2]`

2

Given the marks of a student, return the average mark.

Input to the function: `[70, 85, 90, 55]`
Return from the function: `75.0`

3

Return all numbers that are greater than 50 and less than 100.

Input to the function: `[45, 67, 100, 52, 99, 120, 50]`
Return from the function: `[67, 52, 99]`

4

Return the difference between the largest and the smallest number.

Input to the function: `[14, 3, 27, 9]`
Return from the function: `24`

5

Given marks of a class, return how many students passed. Pass mark is 35.

Input to the function: `[40, 22, 35, 90, 18]`
Return from the function: `3`

6

Return the sum of the numbers at positions 0, 2, 4 and so on.

Input to the function: `[10, 3, 20, 7, 30]`
Return from the function: `60`

7

Given a week of temperatures, return how many days were hotter than the day before.

Input to the function: `[30, 32, 31, 35, 36, 34, 37]`
Return from the function: `4`

8

Return a new array where every odd number is doubled and every even number is halved.

Input to the function: `[4, 7, 10, 3]`
Return from the function: `[2, 14, 5, 6]`

9

Return how many numbers are divisible by both 3 and 5.

Input to the function: `[15, 9, 30, 10, 45, 7]`
Return from the function: `3`

10

Return the number that is closest to the average of the array.

Input to the function: `[2, 9, 4, 15]`
Return from the function: `9`

11

Given daily step counts, return how many days crossed 10,000 steps and the best day's steps, as [days, best].

Input to the function: `[8000, 12000, 10500, 9000, 15000]`
Return from the function: `[3, 15000]`

12

Return the array with every negative number replaced by 0.

Input to the function: `[5, -3, 8, -1, 0]`
Return from the function: `[5, 0, 8, 0, 0]`

13

Return true if the numbers are in increasing order, otherwise false.

Input to the function: `[2, 5, 9, 14]`
Return from the function: `true`

14

Return the sum of the first and the last number.

Input to the function: `[7, 3, 9, 4]`
Return from the function: `11`

15

Given prices of items in a shopping cart, give a 10% discount if the total is above 1000. Return the amount to pay.

Input to the function: `[400, 350, 450]`
Return from the function: `1080`

16

Return how many numbers are palindromes (read the same both ways, like 121).

Input to the function: `[121, 45, 7, 1331, 80]`
Return from the function: `3`

17

Return a new array where each number is replaced by the sum of its digits.

Input to the function: `[123, 45, 9]`
Return from the function: `[6, 9, 9]`

18

Return how many times the sign changes from positive to negative or negative to positive.

Input to the function: `[3, -2, -5, 4, 6, -1]`
Return from the function: `3`

19

Given runs scored in each over of a cricket match, return the over number (starting from 1) with the most runs.

Input to the function: `[6, 12, 4, 15, 9]`
Return from the function: `4`

20

Given Ravi's and Sita's marks in the same five subjects, return in how many subjects Ravi scored more than Sita.

Input to the function: `[78, 65, 90, 55, 82] and [70, 80, 85, 60, 82]`
Return from the function: `2`

21

Given two arrays of the same length, return a new array with the bigger number at each position.

Input to the function: `[4, 9, 2, 7] and [6, 3, 8, 7]`
Return from the function: `[6, 9, 8, 7]`

22

Return the numbers in the first array that are not present in the second array.

Input to the function: `[1, 4, 6, 8, 10] and [4, 8, 12]`
Return from the function: `[1, 6, 10]`

23

Given prices of the same items at Shop A and Shop B, return how much money is saved by buying each item at the cheaper shop instead of buying everything at Shop A.

Input to the function: `[100, 250, 80, 40] and [90, 260, 70, 45]`
Return from the function: `20`

24

Given the scores of three players across matches, return which player (1, 2 or 3) has the highest total.

Input to the function: `[30, 45, 20], [50, 10, 40] and [25, 35, 30]`
Return from the function: `2`

25

Given three arrays, return the average of each array as a new array.

Input to the function: `[10, 20, 30], [5, 15] and [8, 8, 8, 8]`
Return from the function: `[20, 10, 8]`